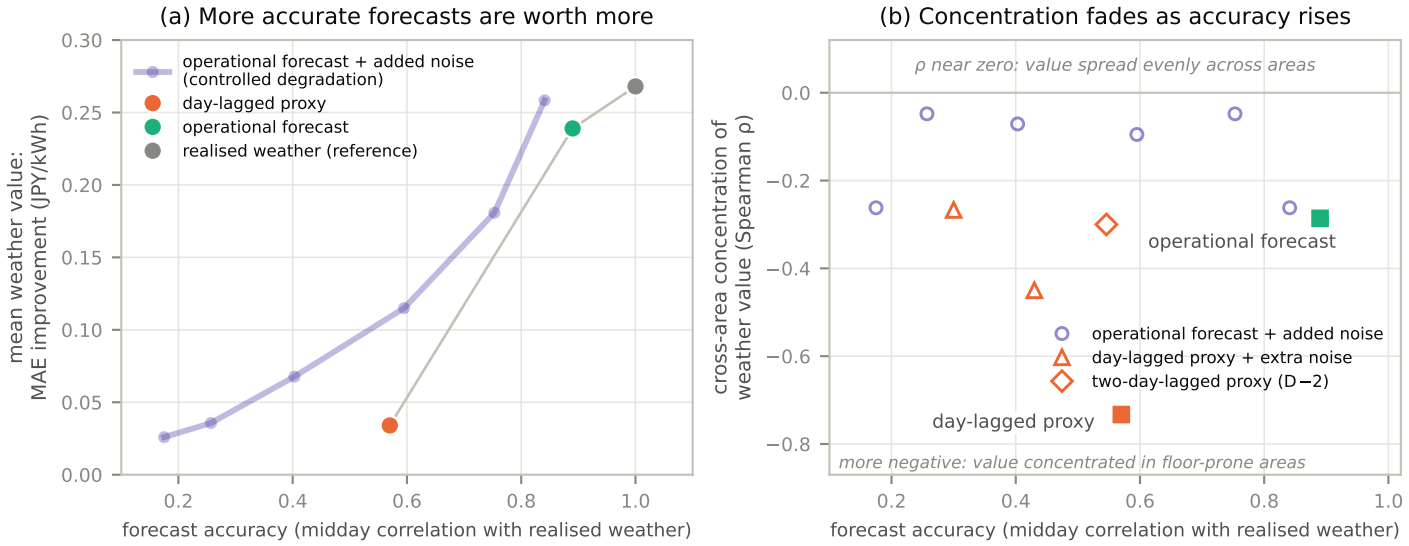
(a) More accurate forecasts are worth more
mean weather value:
MAE improvement (JPY/kWh)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
operational forecast + added noise
(controlled degradation)
day-lagged proxy
operational forecast
realised weather (reference)
0.2
0.4
0.6
0.8
1.0
forecast accuracy (midday correlation with realised weather)
(b) Concentration fades as accuracy rises
ρ near zero: value spread evenly across areas
cross-area concentration of
weather value (Spearman ρ)
0.0
−0.2
−0.4
−0.6
−0.8
operational forecast
operational forecast + added noise
day-lagged proxy + extra noise
two-day-lagged proxy (D−2)
day-lagged proxy
more negative: value concentrated in floor-prone areas
0.2
0.4
0.6
0.8
1.0
forecast accuracy (midday correlation with realised weather)